

WEEKLY PROJECT PROGRESS REPORT (WPPR)

For week commencing 11 April 2026

Programme:	BCA (Artificial Intelligence, Cloud Computing & Devops)
Student Name:	Muhammed Umer Sufaidh
WPPR:	WPPR 06
Register Number:	23BCAICD085
Campus ID:	27079
Project ID:	PRJN26-055
Major Project Title:	Multi-Cloud Hybrid Strategy for Agriculture
College:	YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT
Course:	BCA (Artificial Intelligence, Cloud Computing and DevOps) With IBM & TCS
Semester:	VI Semester
Academic Year:	2023-2026
Internal Guide:	Internal Guide

Targets Set for the Week:

- Implement AWS S3 backup function using boto3
- Test S3 file upload with IAM credentials
- Build the failover controller (run_backup_with_failover)
- Save backup results to last_backup_summary.json
- Run full end-to-end backup test with all three methods

Progress / Achievements for the Week:

1. Implemented backup_aws_s3() function using boto3.client('s3') with IAM credentials, uploading all agriculture files under a timestamped S3 key prefix (agriculture_backup/TIMESTAMP/filename).
2. Built the failover controller run_backup_with_failover() that sequentially attempts Local → Google Drive → AWS S3 backup, logging the result of each step.
3. Implemented the results dictionary and JSON summary export to last_backup_summary.json, storing local, google_drive, aws_s3 boolean results and timestamp.
4. Ran a full end-to-end backup test: all three methods succeeded. Summary JSON confirmed {local: true, google_drive: true, aws_s3: true}.
5. Added a critical alert log message (CRITICAL level) when ALL three backup methods fail, ensuring data-at-risk scenarios are clearly flagged.

Future Work Plans:

- Build the Streamlit web UI (app.py) for user-friendly backup control
- Add buttons for: Create Sample Data + Full Backup, Run Backup Only, Test Local Only
- Add log viewer and summary viewer sections to the UI
- Test the complete application end-to-end via Streamlit interface

Implementation Shown:

Implementation Shown: **Yes**

Remarks by the Internal Guide:

Signature of the Student
Muhammed Umer Sufaidh

Signature of the Internal Guide
Internal Guide